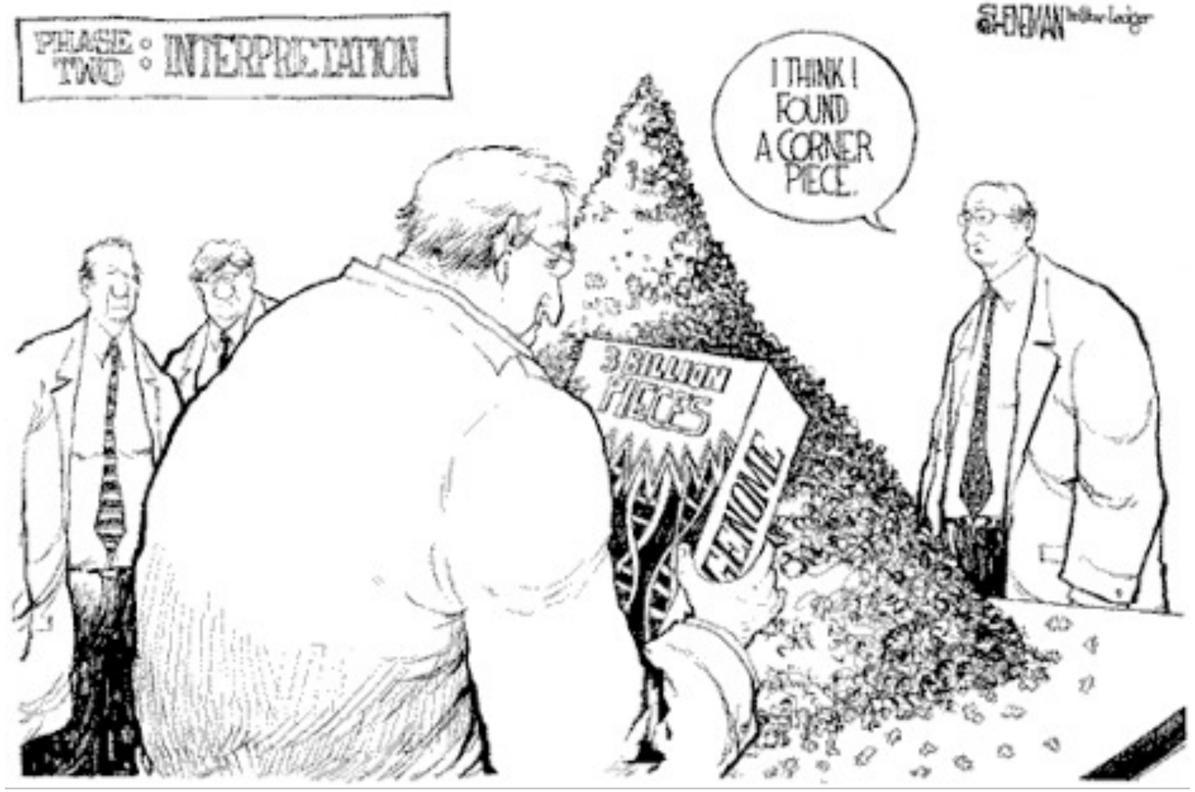
Genome Analysis, Annotation, and Species ID

LaBella Lab
Wild Yeast Project

UNIVERSITY OF NORTH CAROLINA CHARLOTTE

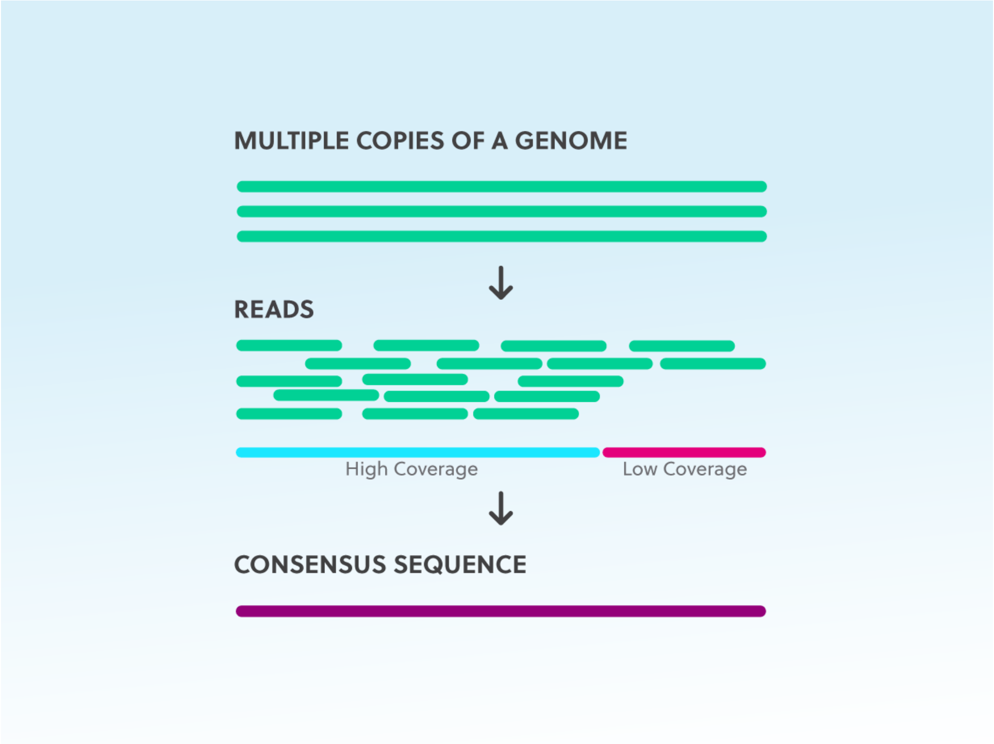
Outline

- Overview of Genome Features, Assembly, & Annotations
- Pipeline Overview
- Activities!



Refresher: DNA Sequencing

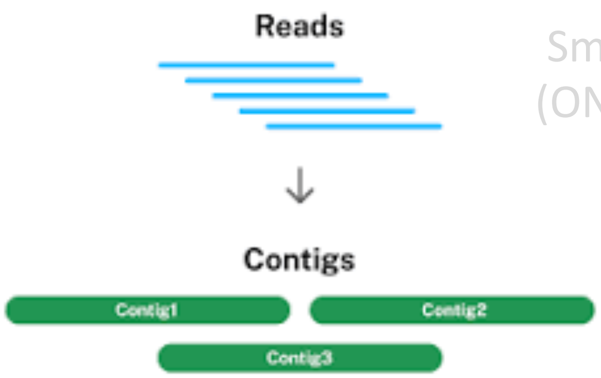
- DNA sequencing technologies sequence multiple copies of DNA
- These are then **assembled** into a single consensus sequence



Genome Assembly

Smallest fragment in sequencing
(ONT can go from 50bp to >4 Mb)

Genome Assembly

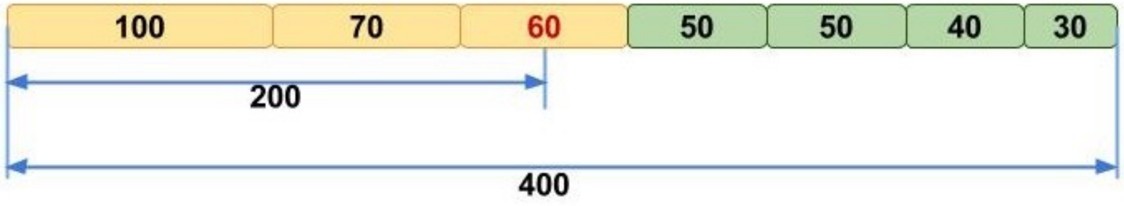


Smallest fragment in sequencing (ONT can go from 50bp to >4 Mb)

Larger fragments assembled based on overlaps in read contents

Quality measured with **N50**

N50: Length of the shortest contig that will cover 50% of the genome assembly when combined with contigs of the same length or longer



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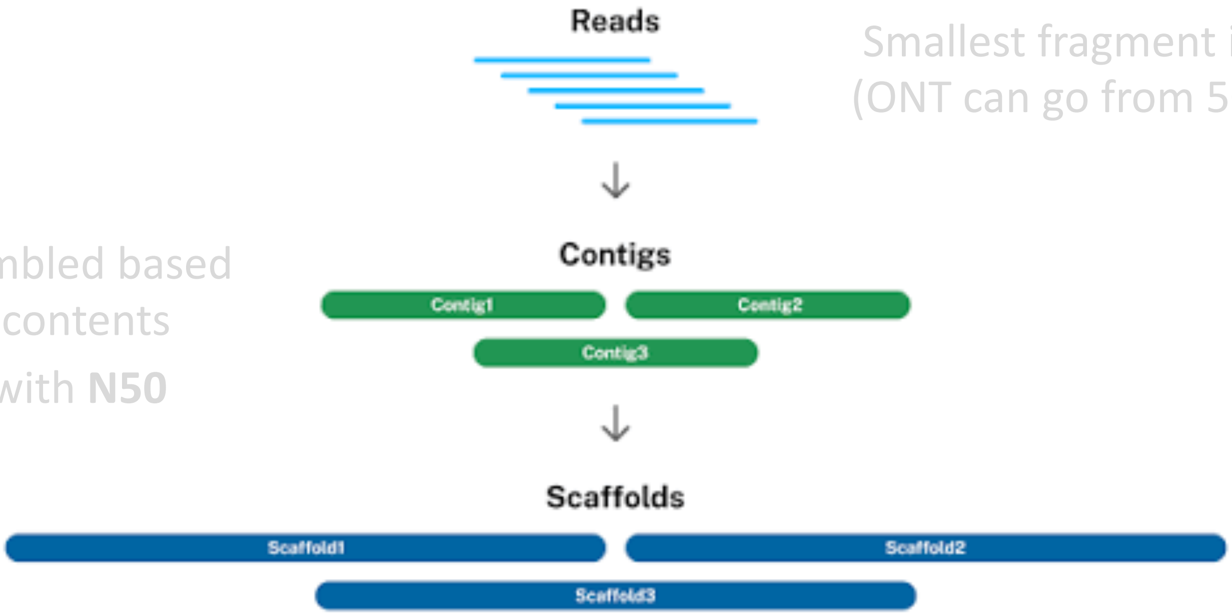
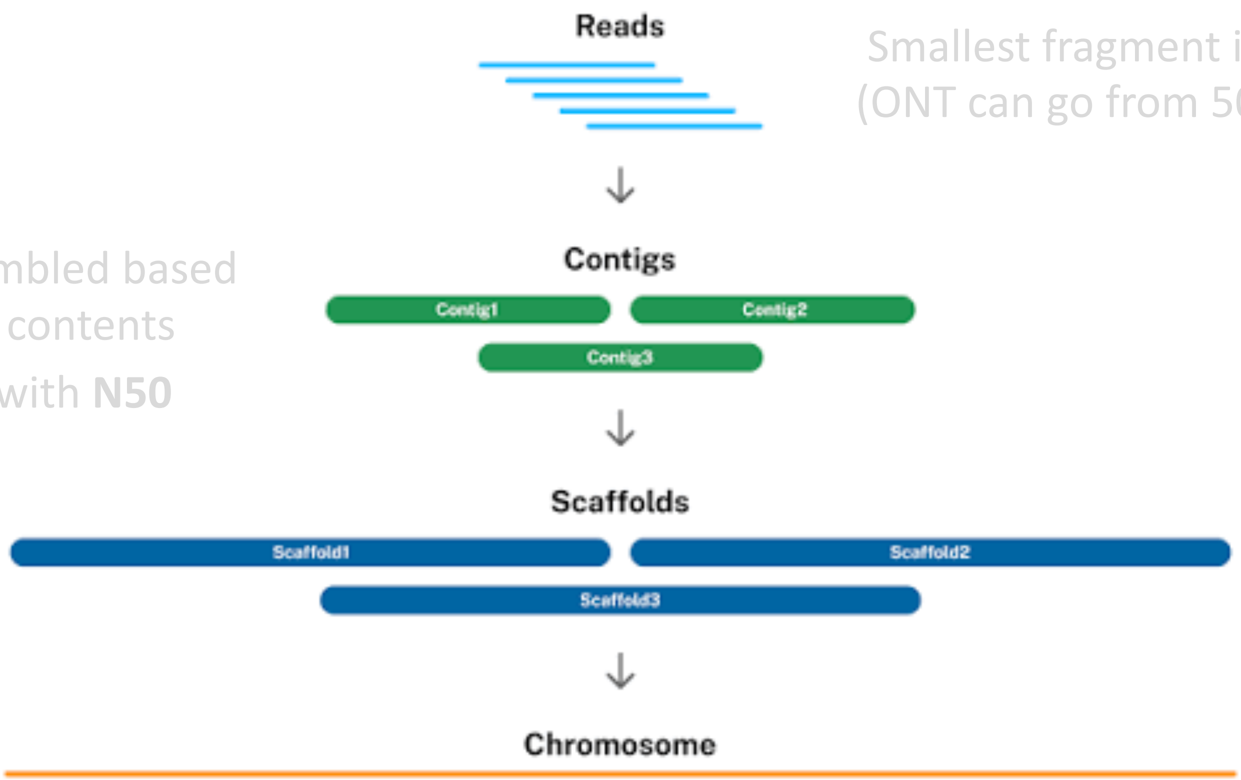


Contigs arranged and linked by position/distance information

Genome Assembly



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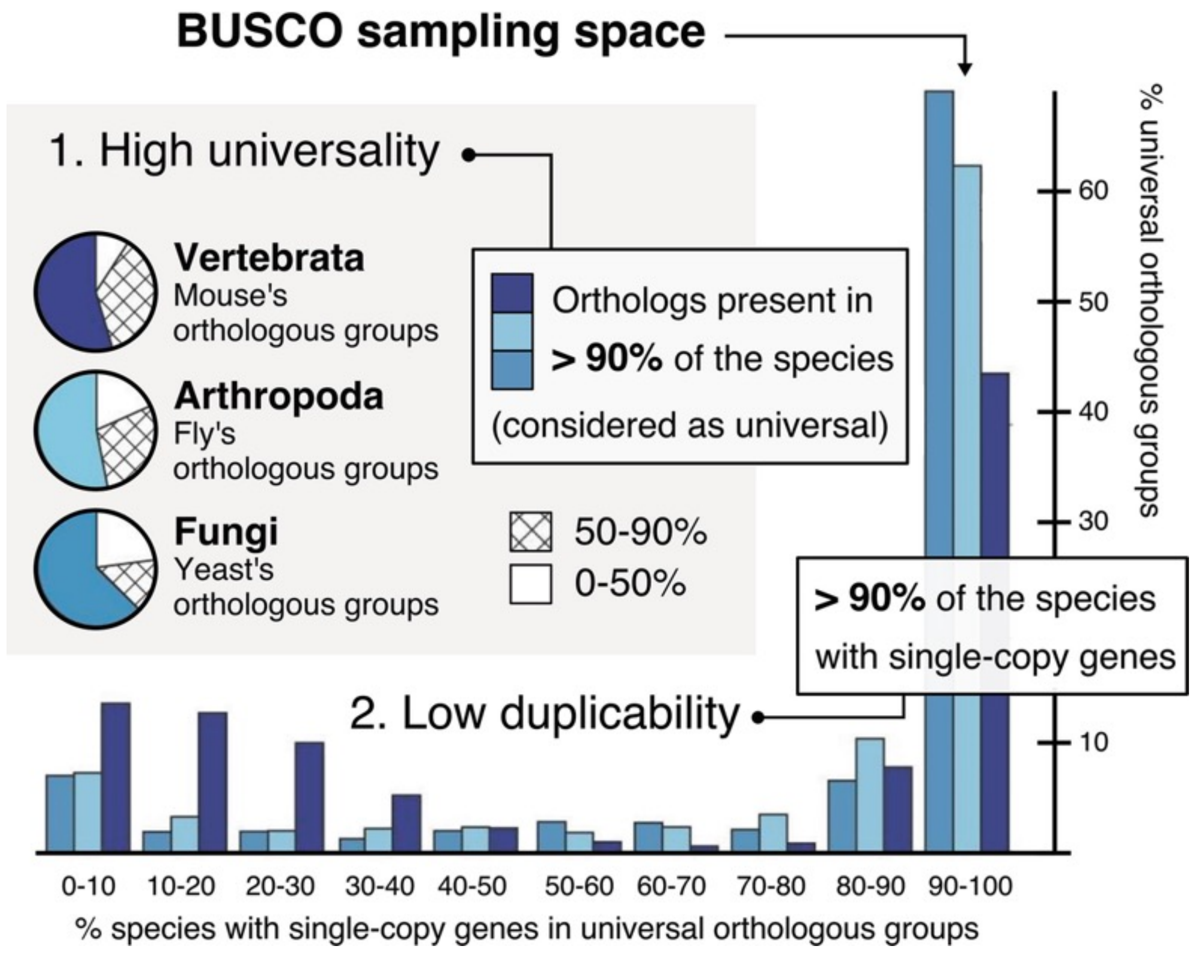
Contigs arranged and linked by position/distance information

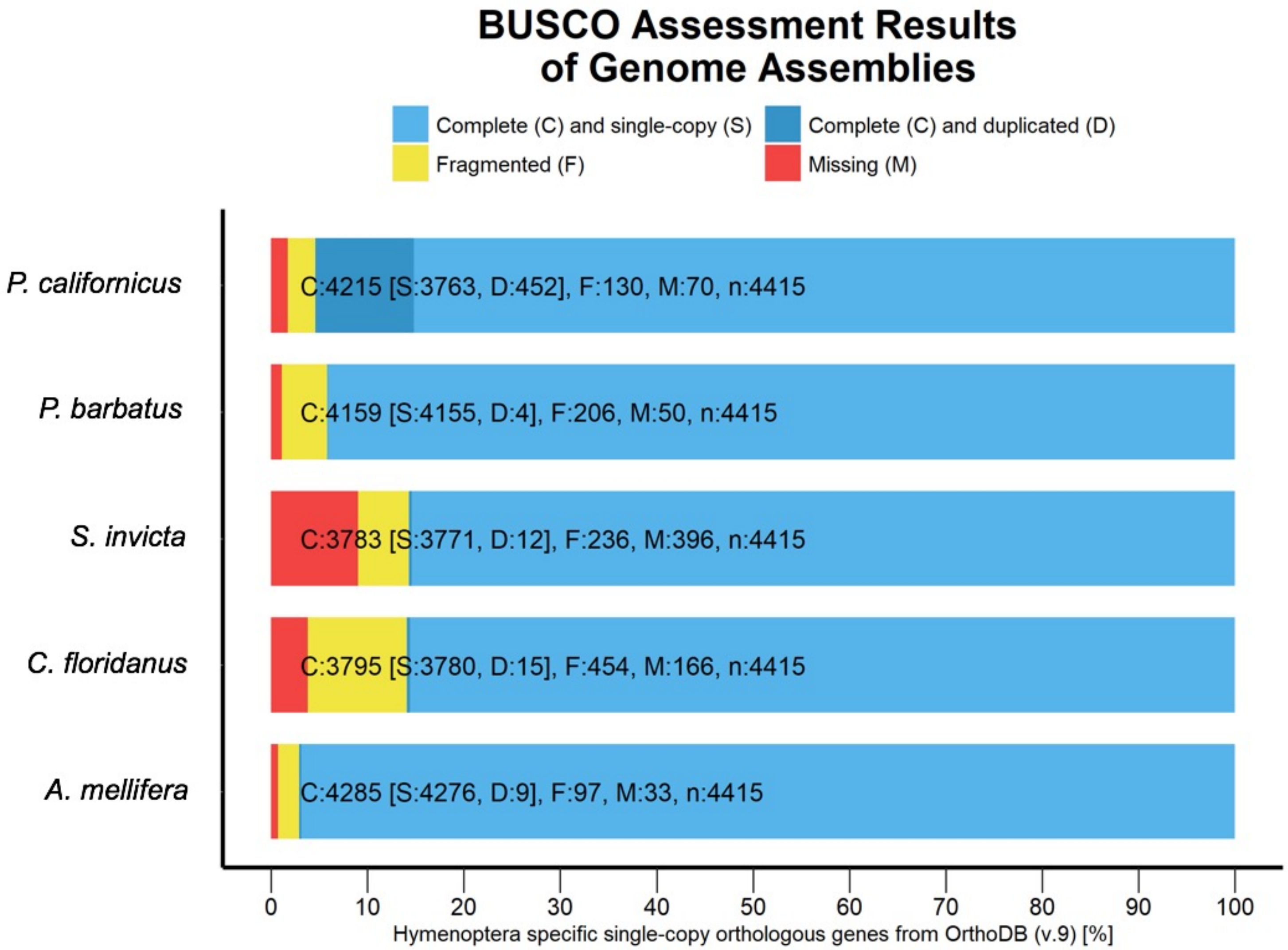
Scaffolds assembled into chromosomes if there's enough mapping information

Genome Assembly Completeness

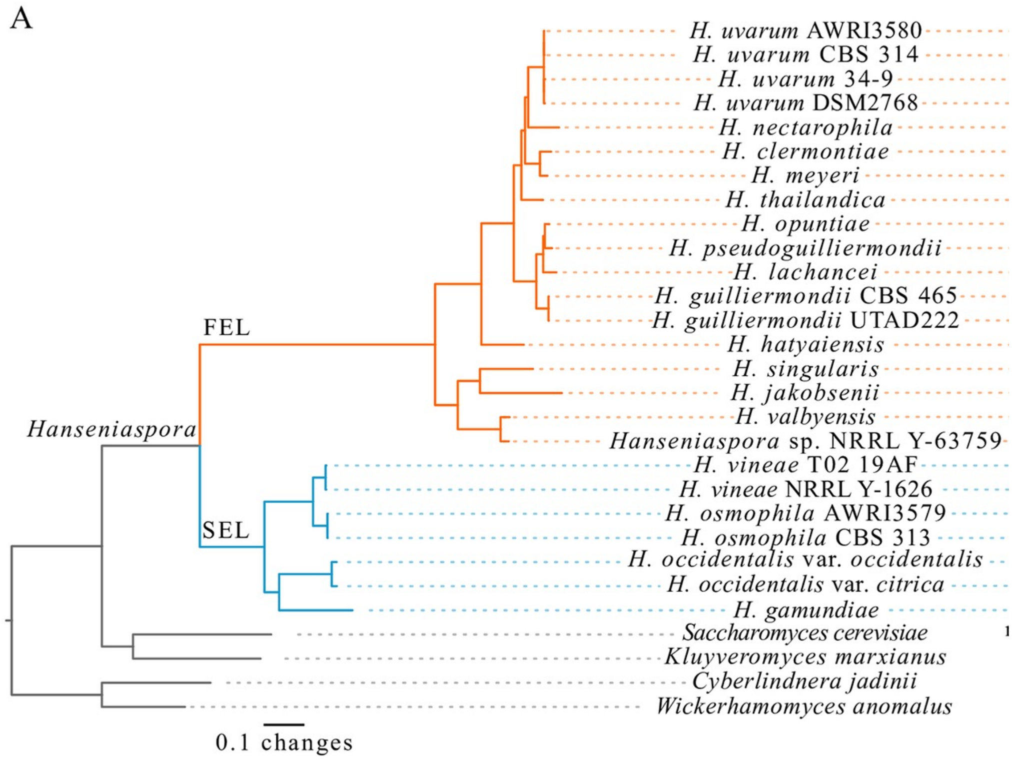
- Complete genome will go from telomere-to-telomere on every chromosome
- Can be assessed by looking for “expected genes” that are conserved across life
- Benchmarking Universal Single-Copy Orthologs (**BUSCO**) score used to assess completeness

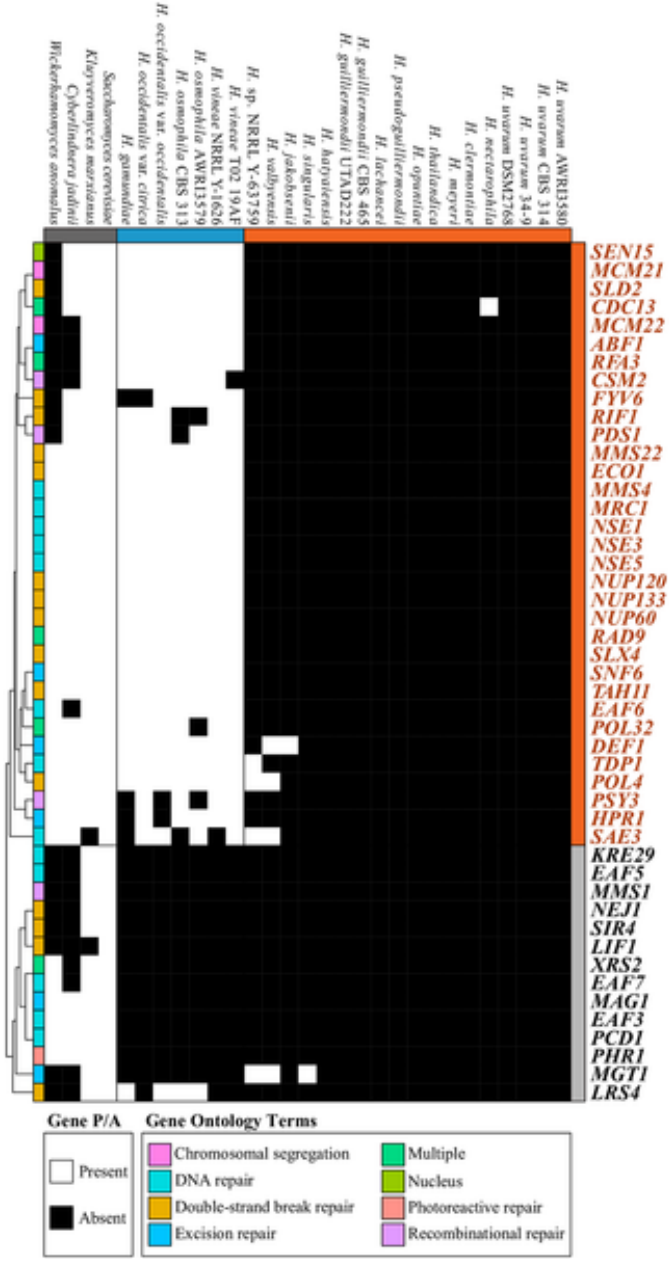


BUSCO Example



BUSCO exceptions….
Because yeasts are very cool.





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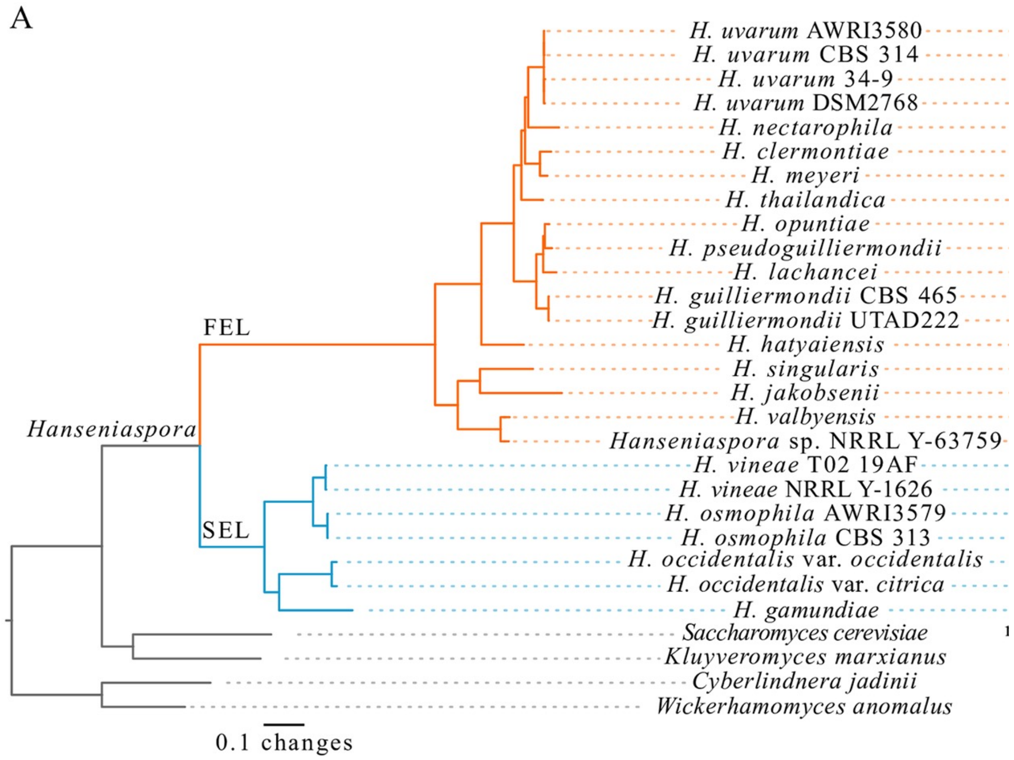
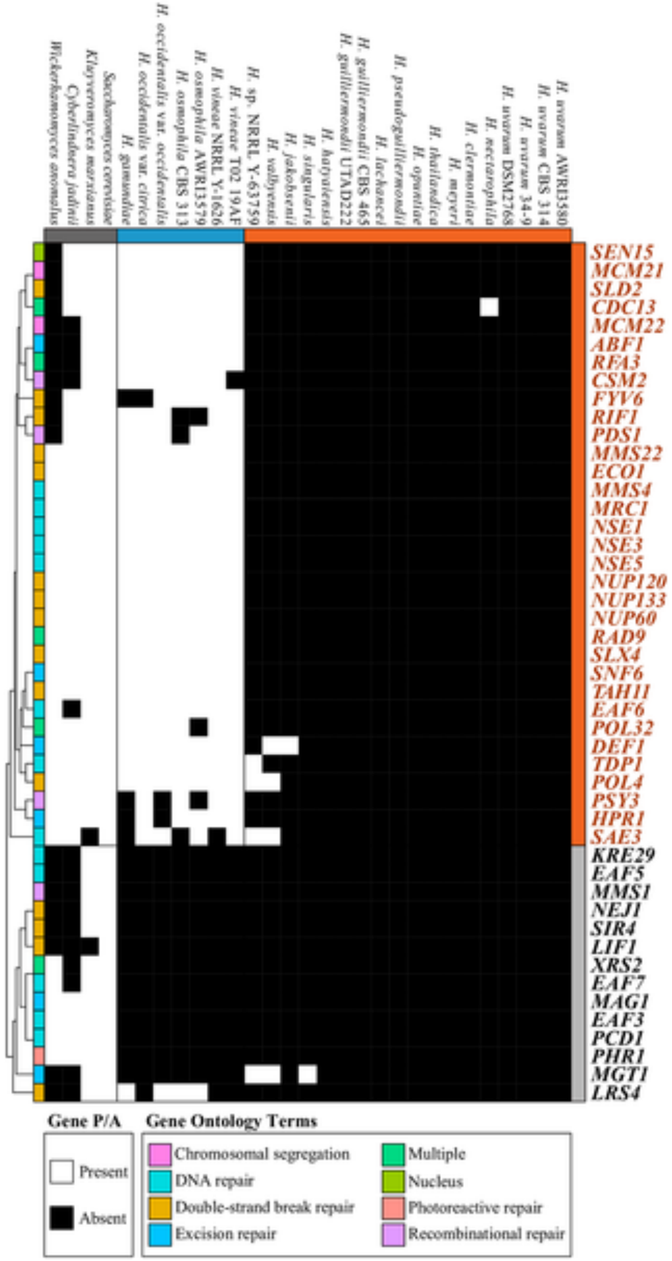
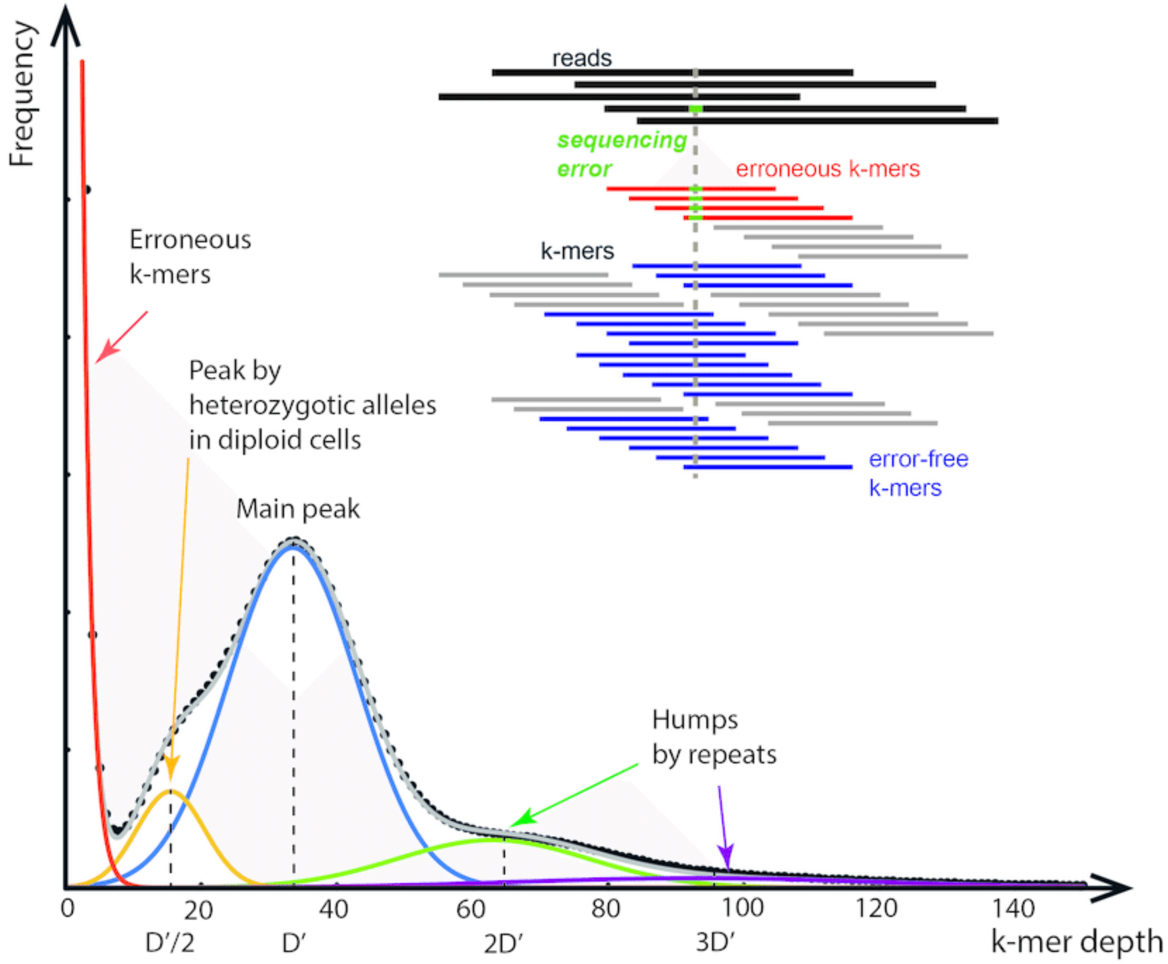




Measuring Correctness

- Challenging to do
- We can identify based on:
  - **Kmer repetition**
  - Read depth



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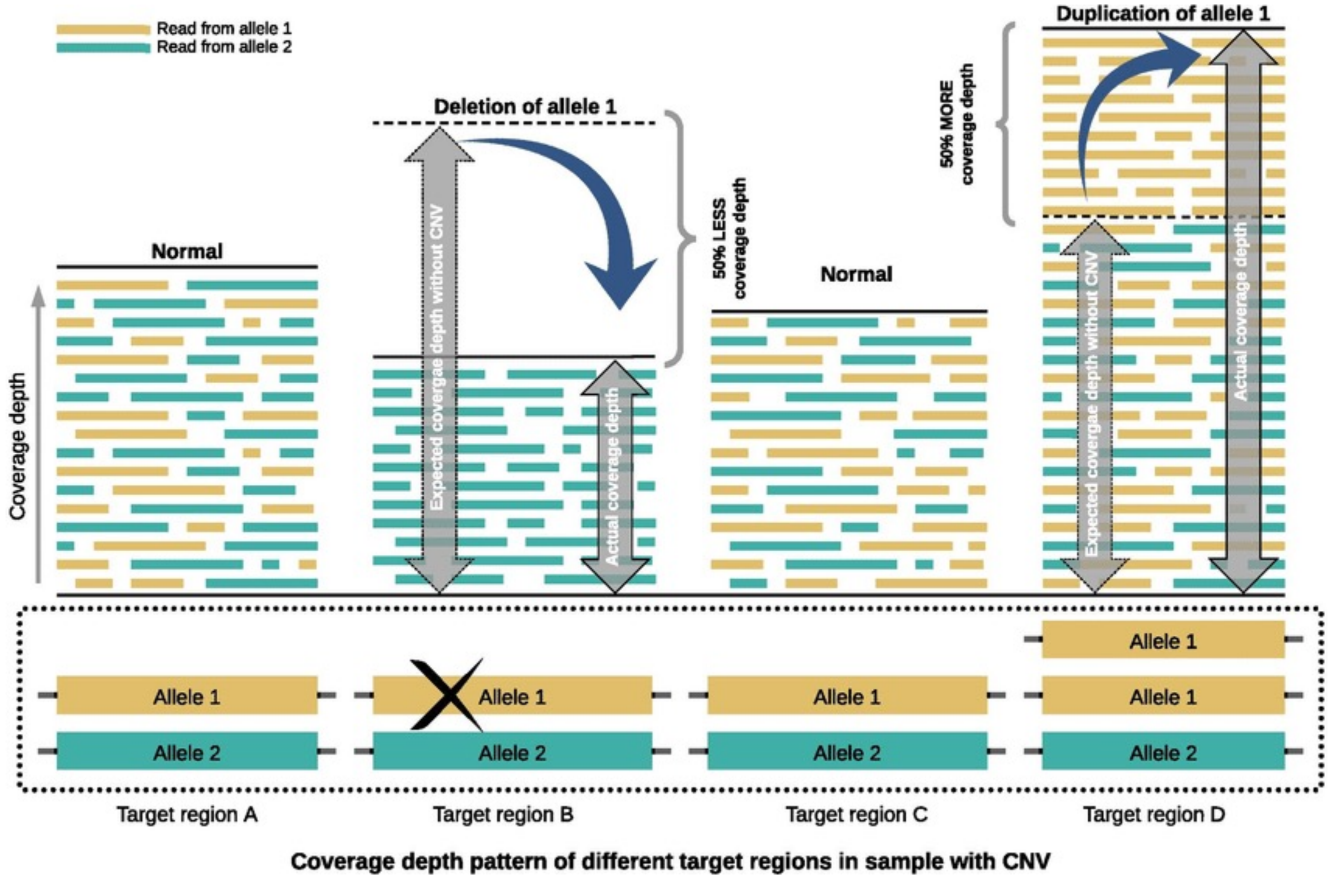

Singh et al 2021

Assembled Genome… now what?

```
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Assembled Genome… now what?

Where are the genes?

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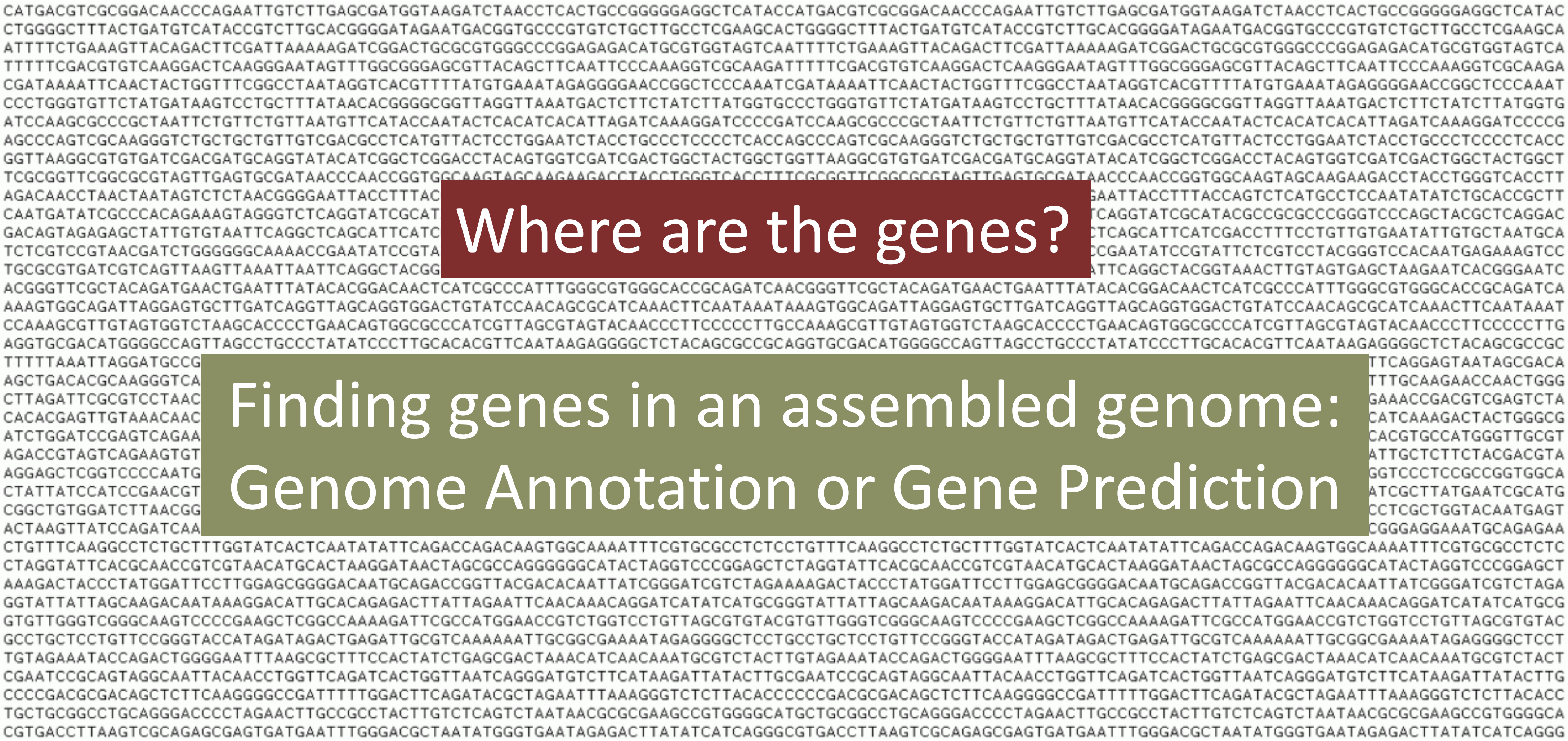


Where are the genes?

Finding genes in an assembled genome: Genome Annotation or Gene Prediction

Gene Annotation

- Genes vary between species, but **eukaryotic genes have conserved features** we can utilize for mathematical models
- Models are trained to some species-specific parameters ➡️ diverse models for gene prediction!

Ab initio Gene Prediction Programs (Possibly with Homology Integration)

Program	Organism	Algorithm*	Website	Homology
GeneID	Vertebrates, plants	DP	http://www1.imim.es/geneid.html	
FGENESH	Human, mouse, Drosophila, rice	HMM	http://www.softberry.com/berry.phtml?topic=fgenesh&group=programs&subgroup=gfind	
GeneParser	Vertebrates	NN	http://beagle.colorado.edu/~eesnyder/GeneParser.html	EST
Genie	Drosophila, human, other	GHMM	http://www.fruitfly.org/seq_tools/genie.html	protein
GenLang	Vertebrates, Drosophila, dicots	Grammar rule	http://www.cbil.upenn.edu/genlang/genlang_home.html	
GENSCAN	Vertebrates, Arabidopsis, maize	GHMM	http://genes.mit.edu/GENSCAN.html	
GlimmerM	Small eukaryotes, Arabidopsis, rice	IMM	http://www.tigr.org/tdb/glimmerm/glmr_form.html	
GRAIL	Human, mouse, Arabidopsis, Drosophila	NN, DP	http://compbio.ornl.gov/Grail-bin/EmptyGrailForm	EST, cDNA
HMMgene	Vertebrates, *C. elegans*	CHMM	http://www.cbs.dtu.dk/services/HMMgene/	
AUGUSTUS	Human, Arabidopsis	IMM,WWAM	http://augustus.gobics.de/	
MZEF	Human, mouse, Arabidopsis, Fission yeast	Quadratic discriminant analysis	http://rulai.cshl.org/tools/genefinder/	

*DP, dynamic programming; NN, neural network; MM, Markov model; HMM, Hidden Markov model; CHMM, class HMM; GHMM, generalized HMM; IMM, interpolated MM.

Wang et al 2004

Funannotate Software

- Designed with fungi in mind
- Combines several different gene prediction models with evidence from transcription and protein data
- Uses UniProtKb/SwissProt database for protein evidence by default

Funannotate

https://github.com/nextgenusfs/funannotate

Genomes are Repetitive

- Multiple kinds of repeating elements (e.g., transposons)
- Repeating elements can do a lot
  - jump into genes and “break” them
  - do nothing
  - make new gene copies
  - carry genes to different regions of the genome or new species
- But need to be **masked** to avoid impact on quality of annotation

ATACTTCAGACGGGGGCCCCCGGGGGAGATACGATACCGAT

RepeatMasker
Uses transposable element library to ID repeats

ATACTTCAGACgggggcccccgggggAGATACGATACCGAT

OR

ATACTTCAGACNNNNNNNNNNNNNNNAGATACGATACCGAT

Finding tRNA genes with tRNAscan SE

Species	Total tRNAs	tRNA types (62 possible)
Humans	416	48
Mouse	401	47
Arabidopsis	580	47
E. coli	89	42
H. volcanii	53	46

Finding tRNA genes with tRNAscan SE

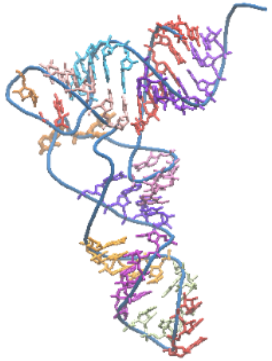
tRNAscan-SE

Searching for tRNA genes in genomic sequence

This web server utilizes tRNAscan-SE 2.0

When using this web server, please cite one of the references.

Lowe, T.M. and Chan, P.P. (2016) tRNAscan-SE On-line: Search and Contextual Analysis of Transfer RNA Genes. *Nucl. Acids Res.* **44**:W54-57.
Chan, P.P. and Lowe, T. M. (2019) tRNAscan-SE: Searching for tRNA Genes in Genomic Sequences. *Methods Mol Biol.* **1962**:1-14.



When using the downloadable version of tRNAscan-SE 2.0 (below), please cite

Chan, P.P., Lin, B.Y., Mak, A.J., and Lowe, T.M. (2021) tRNAscan-SE 2.0: Improved Detection and Functional Classification of Transfer RNA Genes. *Nucl. Acids Res.* **49**:9077–9096.

Download

Source code: tRNAscan-SE-2.0.12.tar.gz
Change Log

tRNAscan-SE covariance model seed alignments (Stockholm format): tRNAscan-SE_CM_alignments.tar.gz

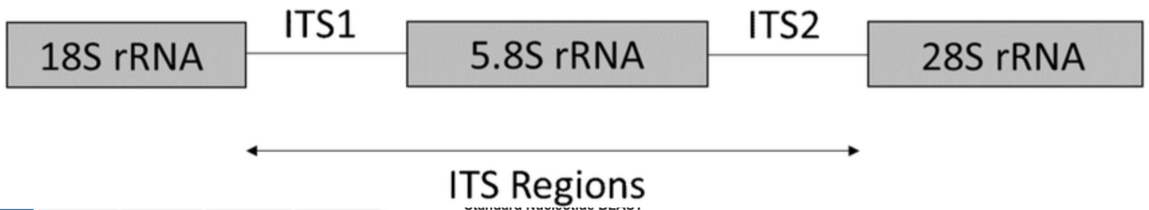
Example tRNA sequences

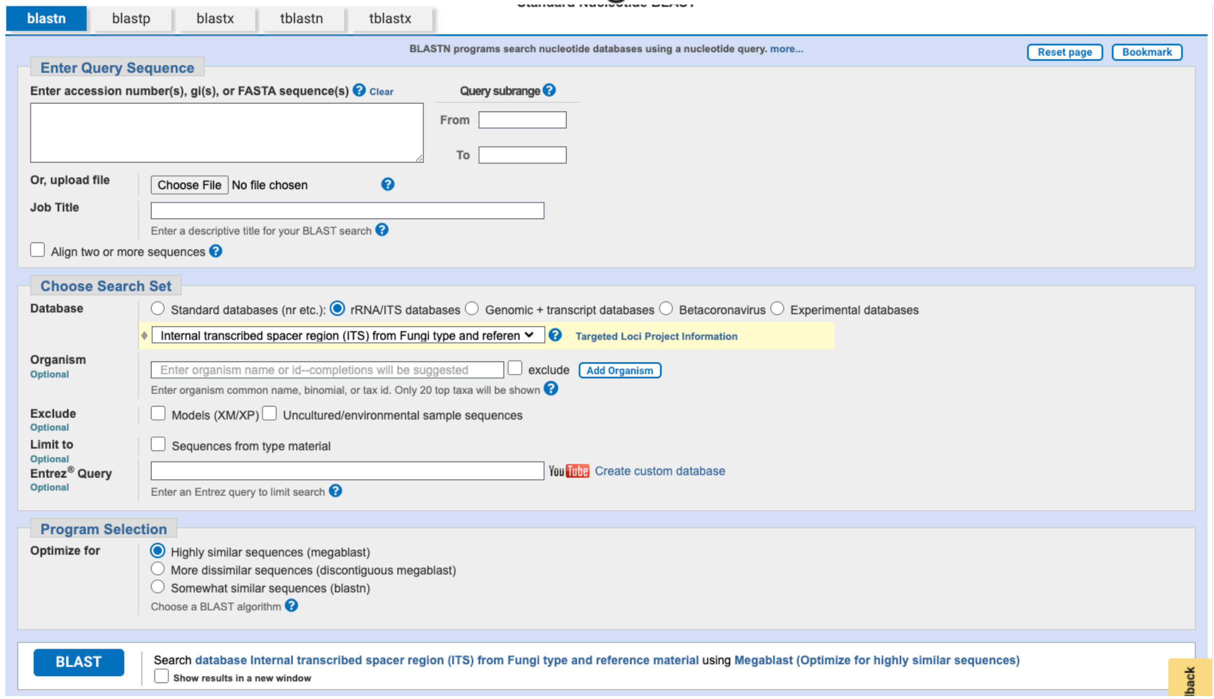
Search options

Sequence source	Eukaryotic
Search mode	Default

Fungal Species ID

- Internal Transcribed Spacer (ITS) used for fungal DNA barcoding/identification
- Consists of 2 parts (ITS1, ITS2) in highly conserved regions of the genome (ribosome genes)
- Region can accumulate mutations freely without risk of being lost
- These mutations are how we tell species apart





Activities

- Genome Annotation
  - https://github.com/The-Lab-LaBella/Wild_Yeast_Summer_2026/blob/main/Genome_Annotation.md
- Species ID (*do this while you wait for annotation to finish!*
  - https://github.com/The-Lab-LaBella/Wild_Yeast_Summer_2026/blob/main/Species_ID.md

Record results in the Google Sheet!

An aside on contamination

- Contamination can happen at multiple stages
- Can impact downstream analyses and conclusions
- Ward and Scopel et al 2026 (G3) go over potential consequences and solutions (B-allele frequency plots)